CHANDRAYAAN-2 · OHRC · LUNAR PSR

LunaPSR

Making the Moon’s permanent shadows analyzable — without inventing detail.

A read-only lab that enhances faint Chandrayaan-2 OHRC images, compares methods, and records exactly what changed. For lunar scientists, image analysts, and reviewers.

Read-only source | Model-based confidence | Ground-truth tested

SOURCE SIGNAL	MODELED NOISE	PRODUCT SCALE
1–5 DN	0.511 DN	1.2 Gpx

All enhanced imagery is a **derived representation**. We do not claim recovered geology, measured SNR, or information the sensor did not collect.

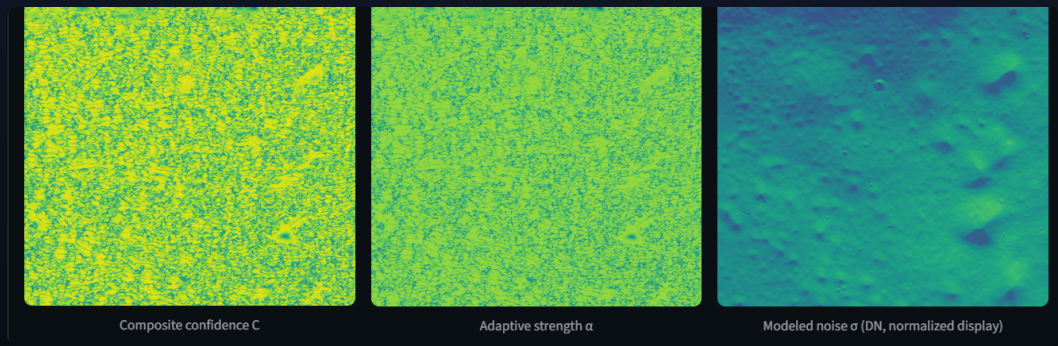
01 · THE PROBLEM IT SOLVES

The signal is barely above the noise.

Permanently Shadowed Regions (PSRs) near the Moon’s poles receive no direct sunlight. OHRC, the Orbiter High Resolution Camera, can image them — but the data are extremely faint, and very easy to over-enhance.

- **Weak signal:** raw values span **1–5 DN** in the studied region (mean 3.056, std 0.467). DN is the raw brightness unit.
- **Noise:** the Phase 4B physics model puts total noise at **σ ≈ 0.511 DN** — a **model-based SNR ≈ 6**, not a measured SNR.
- **False detail:** a global stretch mostly reveals sensor noise. Aggressive Retinex can create edges, textures, and isolated components that were not in the data.
- **No audit trail:** comparing methods on a 1.2-gigapixel product is slow, inconsistent, and hard to reproduce.

> The real problem: an enhanced image can look better while becoming less trustworthy.



Model-based confidence, adaptive strength, and modeled noise — diagnostics, not measurements.

Weak contrast · Noise amplification · False detail · No ground truth · No audit trail

Source: README §2, §12; docs/phase5a_ssr_investigation.md

MODEL-BASED · NOT MEASURED

One workflow: from raw pixels to evidence you can review.

LunaPSR is not a single filter. It is a complete lab for reading, enhancing, comparing, and documenting real lunar images — without modifying the source product.

Real OHRC product → Mask & normalize → Noise model → Enhance → Diagnostics & record

The pipeline is always the same: read-only in, documented evidence out.

READ-ONLY
1.2-gigapixel product, memory-mapped; source never changed.

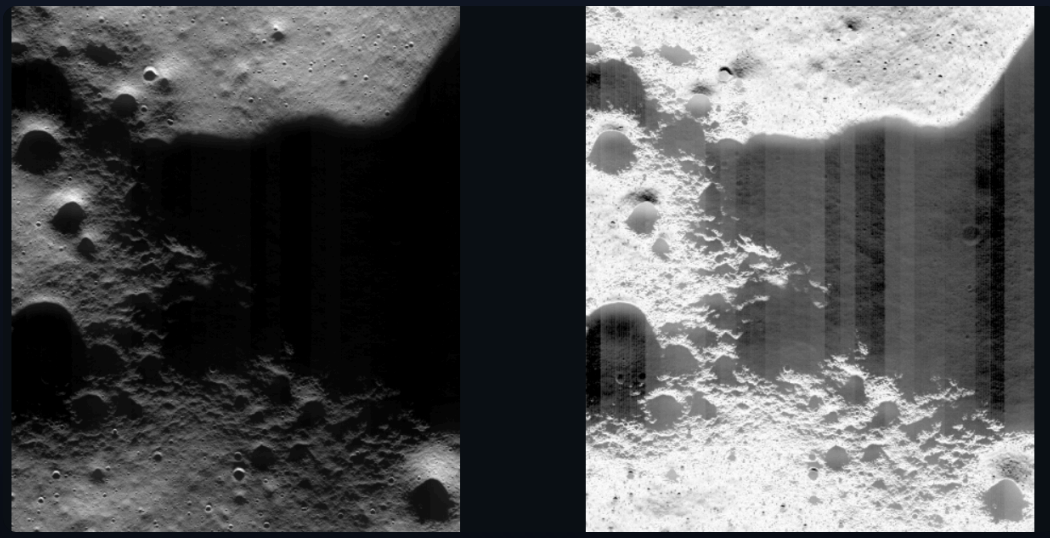
NOISE-AWARE
Model-based SNR and confidence, not ad-hoc contrast.

MULTI-METHOD
Gamma, CLAHE, SSR, denoisers, LunaPSR, Adaptive Retinex.

DIAGNOSTICS
Noise, confidence, structure, and edge-preservation maps.

AUDITABLE
Metrics, processing history, JSON / CSV / PNG exports.

USABLE
9-tab Streamlit lab; 304 tests keep the workflow honest.



The lab: documented controls, Original / SSR / LunaPSR views, and headline diagnostics. Every panel is a derived visualization.

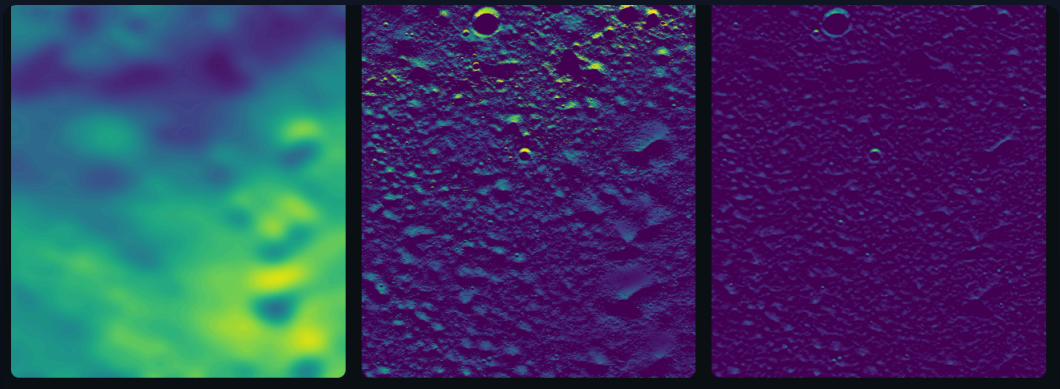
Source: README §4–§6; app.py; core/ohrc/

Enhance only where the signal can support it.

A normal enhancement applies one curve everywhere. LunaPSR uses a physics-based noise model to decide how much enhancement each part of the image can justify.

$$\sigma(DN) = \sqrt{\left(DN/g + (\sigma_{read}/g)^2 + 1/12 \right)}$$

- **Noise is predicted, not guessed.** The model combines shot noise, read noise, and quantization noise at each brightness level.
- **Two-factor confidence:** is the local signal strong, and is local variation above the modeled noise?
- **Adaptive strength α** stays conservative where the data look like noise, and increases only where real structure is supported.
- **Adaptive Retinex** separates illumination, response, and structure instead of applying one global curve to the whole scene.



Illumination, robust Retinex response, and medium-scale structure used as variation evidence. Gain *g* is inferred from metadata — not an official calibration claim.

> In plain terms: the noise model is the gate. Where the data cannot support detail, LunaPSR holds back.

Phase 5B prototype — validation ongoing

MODEL-BASED SNR

Source: docs/phase4b_noise_model.md; docs/phase5b_lunapsr_retinex.md

We tested it against known ground truth — and report where it loses.

Phase 5C injects known craters and boulders into eight real 256×256 OHRC dark tiles. Five methods, one generic detector, per-tile 5% false-positive calibration.

0.882	0.868	0.704	0.638
Raw AUC — best overall	LunaPSR AUC	LunaPSR boulder AUC ≤ 0.5 DN	Raw boulder AUC ≤ 0.5 DN

- **Raw wins overall** (AUC 0.882 vs 0.868). Enhancement is not required to detect these features.
- **LunaPSR has a narrow win.** At ≤ 0.5 DN it ties raw overall (0.744 vs 0.742) and leads on faint 12–20 px boulders (0.704 vs 0.638; SSR 0.559).
- **It also loses.** LunaPSR is the worst method for 6 px boulders and less sensitive than raw for faint craters. SSR amplifies noise **~50×**; LunaPSR **~14.5×**.

> The win is narrow, measurable, and honest — not a clean sweep. AUC measures how well a method separates real features from background: 0.5 is chance, 1.0 is perfect.



Detection versus injected contrast for boulders. The hard regime is low input SNR; the curves show the size-dependent range where LunaPSR helps.

Source: docs/phase5c_ground_truth_benchmark.md; outputs/phase5c/

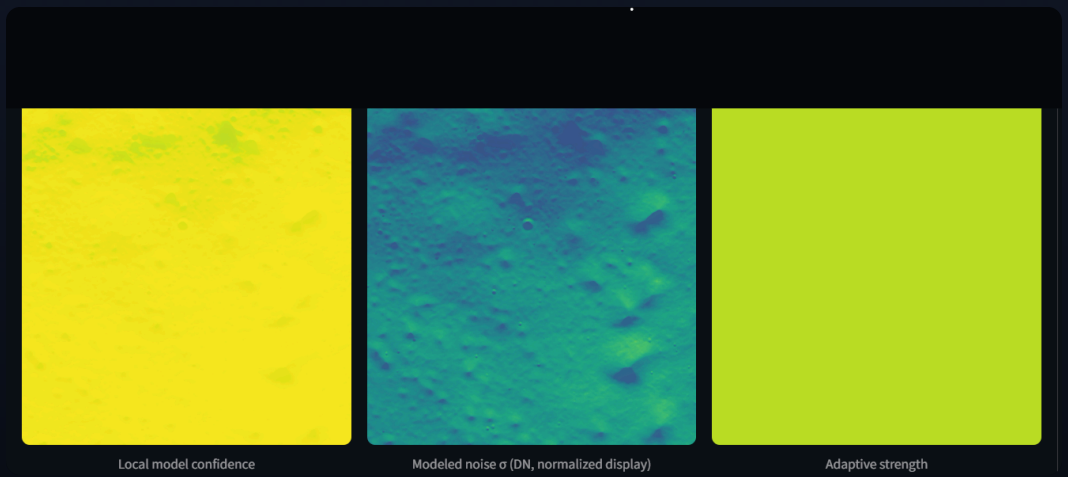
What you can do with LunaPSR.

Use it to inspect a real product, compare enhancement methods, and see exactly what each step changed — before you trust the result.

- **Inspect** a real OHRC product read-only.
- **Compare** gamma, CLAHE, SSR, denoisers, LunaPSR, and Adaptive Retinex side by side.
- **See** confidence, noise, structure, and edge diagnostics for every view.
- **Export** derived images, metrics, and a reproducible run record.
- **Run the lab** locally with the full 1.2 Gpx product; the cloud demo uses the bundled sample.

Phase 5C complete | Phase 5B prototype | Demo private

> The goal is not a prettier image. It is a more trustworthy analysis.



Diagnostic maps: local model-based confidence, modeled noise σ, and adaptive strength. Research diagnostics on derived data.